# Global Economic Shifts

ECON-220

Gains and Losses from Trade in the Specific-Factors Model

Instructor: Muhebullah Karimzada

# Questions to Consider

1. Do you personally gain from inexpensive imported goods?
2. Besides you, who gains and who loses from trade?
3. What government policies can help firms and workers that lose from trade?

# Introduction

- The argument from the Ricardian model that trade generates gains for *all* workers is too simple because labor is the only factor of production in that model.
- We relax that assumption with the specific-factors model, where land can be used only in the agriculture sector and capital can be used only in the manufacturing sector; labor is used in both sectors.
- From the Ricardian model, we learned that free trade affects relative prices, and this in turn affects the earnings of factors of production.
- The question addressed by the specific-factors model is how trade, through changes in *relative prices,* affects the earnings of labor, land, and capital.

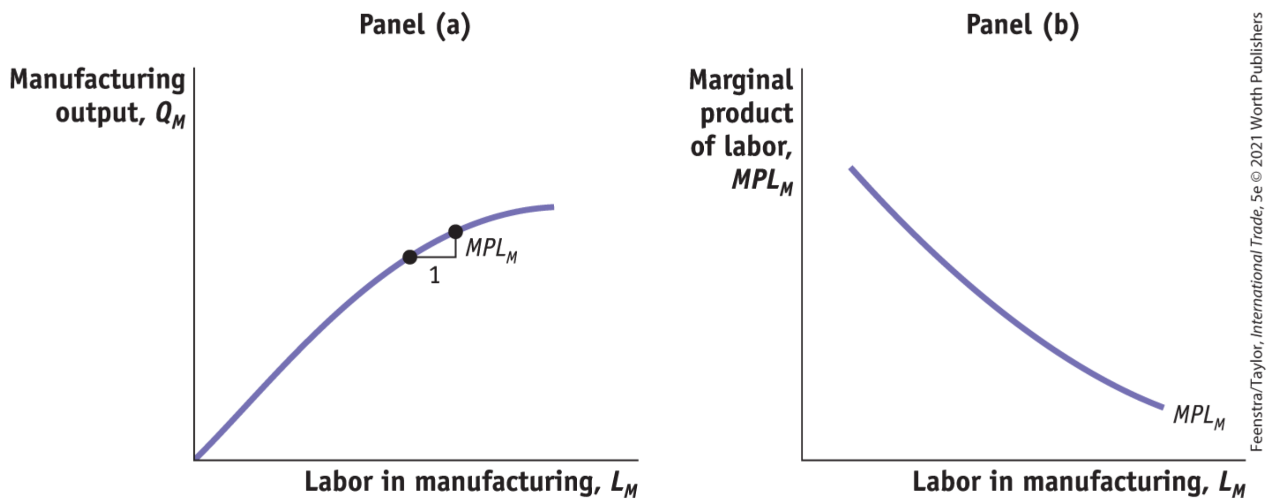
# 1) Specific-Factors Model (part 1)

- The specific-factors model we will develop has the following features:
  - Once again there are two countries: Home and Foreign.
  - Manufacturing uses labor and capital, and agriculture uses labor and land.
  - In each industry, increases in the amount of labor used are subject to **diminishing returns;** that is, the marginal product of labor declines as the amount of labor used in the industry increases.
  - For now let’s focus on the Home country.

# 1) Specific-Factors Model (part 2)

- The Home Country

FIGURE 3-1



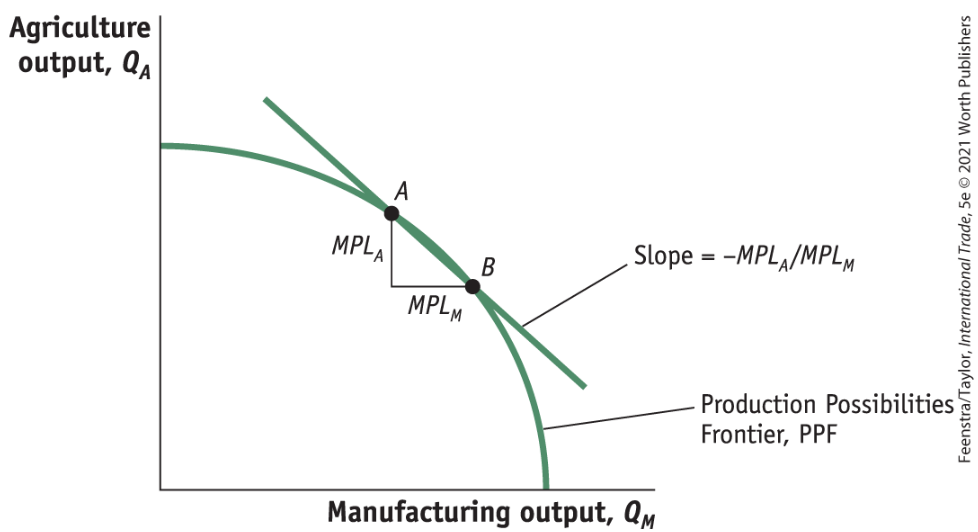

Panel (a) Manufacturing Output As more labor is used, manufacturing output increases, but it does so at a diminishing rate.

Panel (b) Diminishing Marginal Product of Labor An increase in the amount of labor used in manufacturing lowers the marginal product of labor.

# 1) Specific-Factors Model (part 3)

- The Home Country
- Production Possibilities Frontier

FIGURE 3-2




Production Possibilities Frontier

The production possibilities frontier shows the amount of agricultural and manufacturing outputs that can be produced in the economy with labor.

# 1) Specific-Factors Model (part 4)

- The Home Country
- Opportunity Cost and Prices
  - As in the Ricardian model, the slope of the PPF equals the opportunity cost or relative price of the good on the horizontal axis; here it is manufacturing.
  - Firms hire labor up to the point where the cost of one more hour of labor (the wage) equals the value of one more hour of labor in production.

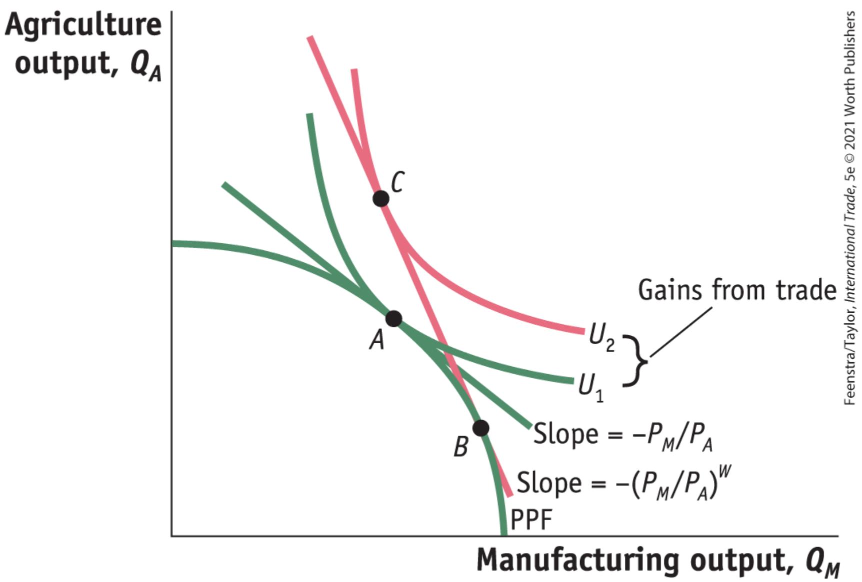
$$W = P_M \cdot MPL_M$$

$$W = P_A \cdot MPL_A$$

# 1) Specific-Factors Model (part 5)

- The Home Country
- Opportunity Cost and Prices

FIGURE 3-3




In the absence of international trade, the economy produces and consumes at point *A*.

The relative price of manufactures, $P_M/P_A$, is the slope of the line tangent to the PPF and indifference curve $U_1$, at point *A*.

With international trade, the economy is able to produce at point *B* and consume at point *C*.

The world relative price of manufactures, $(P_M/P_A)^W$, is the slope of the line *BC*.

The rise in utility from $U_1$ to $U_2$ is a measure of the gains from trade for the economy.

# 1) Specific-Factors Model (part 6)

- The Foreign Country
  - Let us assume that the Home no-trade relative price of manufacturing is lower than the Foreign relative price.

$$(P_M/P_A) < (P^*_M/P^*_A)$$

  - This means that Home can produce manufactured goods relatively cheaper than Foreign.
  - Put another way, Home has a comparative advantage in manufacturing.

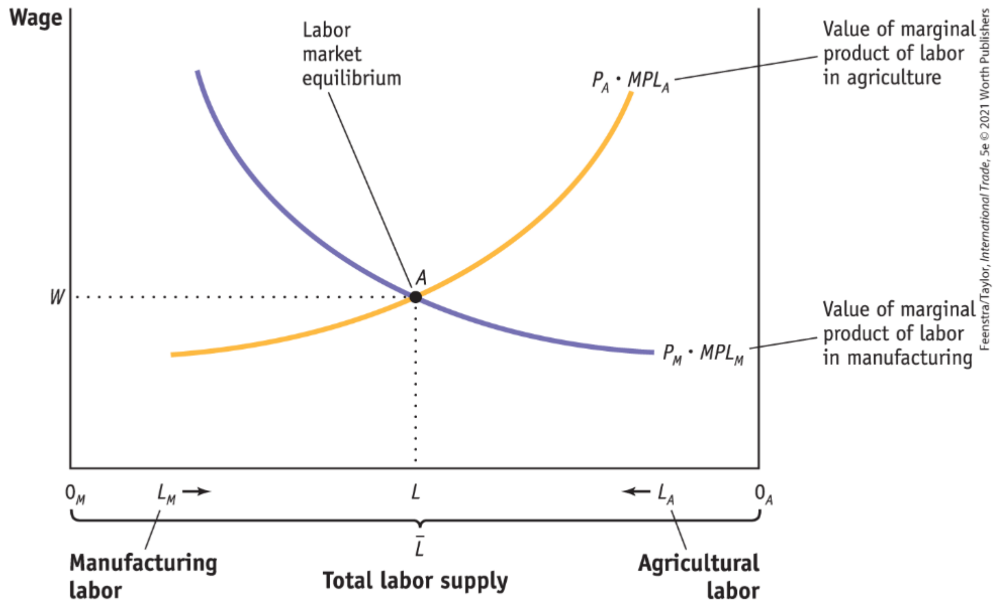
# 1) Specific-Factors Model (part 7)

- Overall Gains from Trade
  - The good whose relative price goes up (manufacturing, for Home) is exported.
  - The good whose relative price goes down (agriculture, for Home) is imported.
  - By exporting manufactured goods at a higher price and importing food at a lower price, Home is better off than it was in the absence of trade.

# 2) Earnings of Labor (part 1)

- Determination of Wages

FIGURE 3-4 (1 of 2)

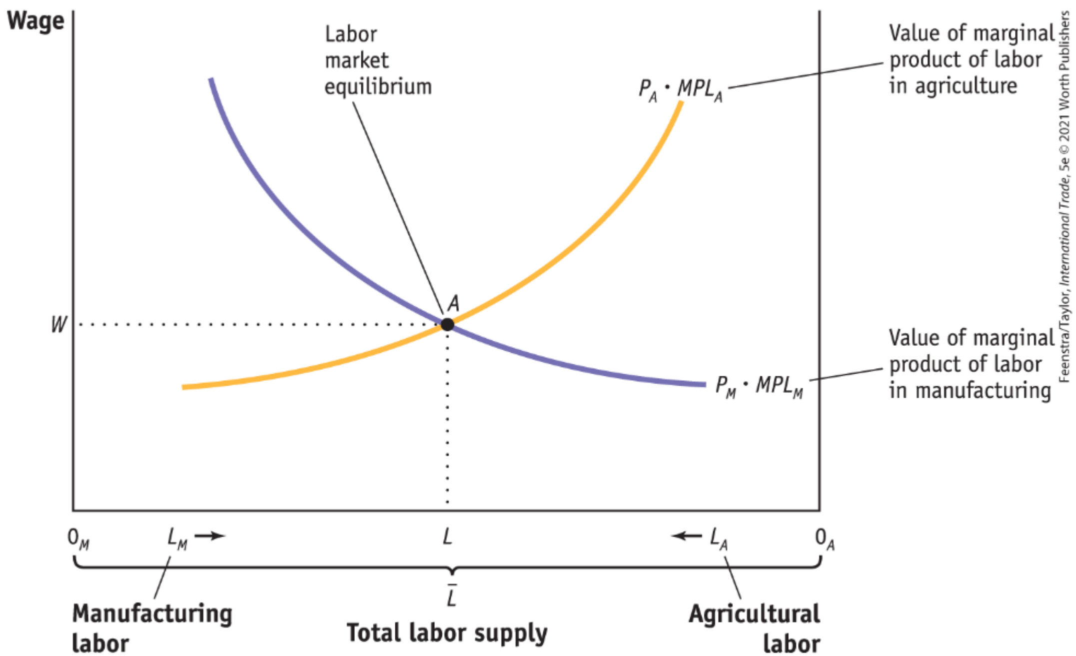



The amount of labor used in manufacturing is measured from left to right along the horizontal axis, and the amount of labor used in agriculture is measured from right to left.

# 2) Earnings of Labor (part 2)

- Determination of Wages

FIGURE 3-4 (2 of 2)




Labor market equilibrium is at point *A*. At the equilibrium wage of *W*, manufacturing uses $0_M L$ units of labor and agriculture uses $0_A L$ units.

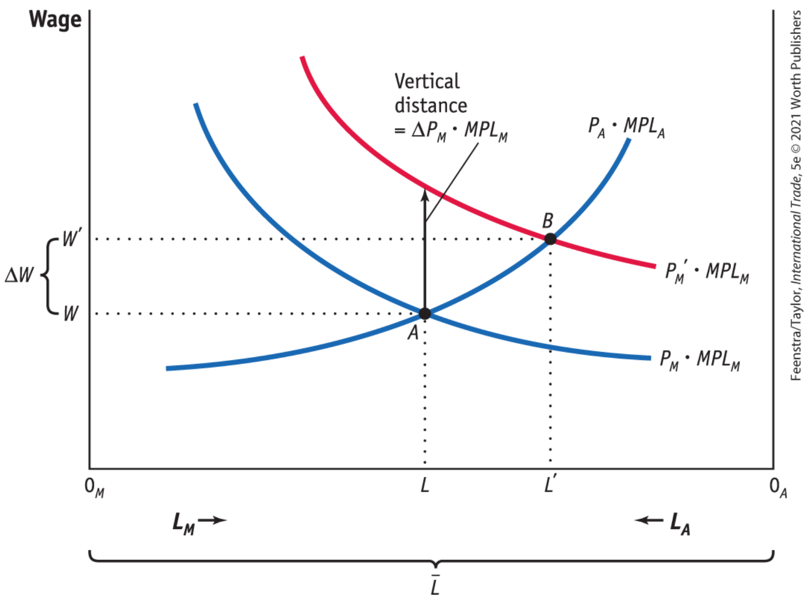
# 2) Earnings of Labor (part 3)

- Change in Relative Price of Manufactures
- Now consider an increase in the price of the manufactured good ($P_M$).
  - With an increase in the price of the manufactured good the curve $P_M \bullet MPL_M$ shifts up.
  - Therefore, the labor used in manufacturing rises, and labor used in agriculture falls.
  - The wages also increase, but this increase is less than the upward shift $\Delta P_M \cdot MPL_M$

# 2) Earnings of Labor (part 4)

- Change in Relative Price of Manufactures

FIGURE 3-5




Increase in the Price of Manufactured Goods

With an increase in the price of the manufactured good, the curve $P_M \cdot MPL_M$ shifts up to $P_M' \cdot MPL_M$ and the equilibrium shifts from point *A* to *B*.

The labor used in manufacturing rises from $0_M L$ to $0_M L'$, and labor used in agriculture falls from $0_A L$ to $0_A L'$.

The wage increases from $W$ to $W'$, but this increase is less than the upward shift $\Delta P_M \cdot MPL_M$.

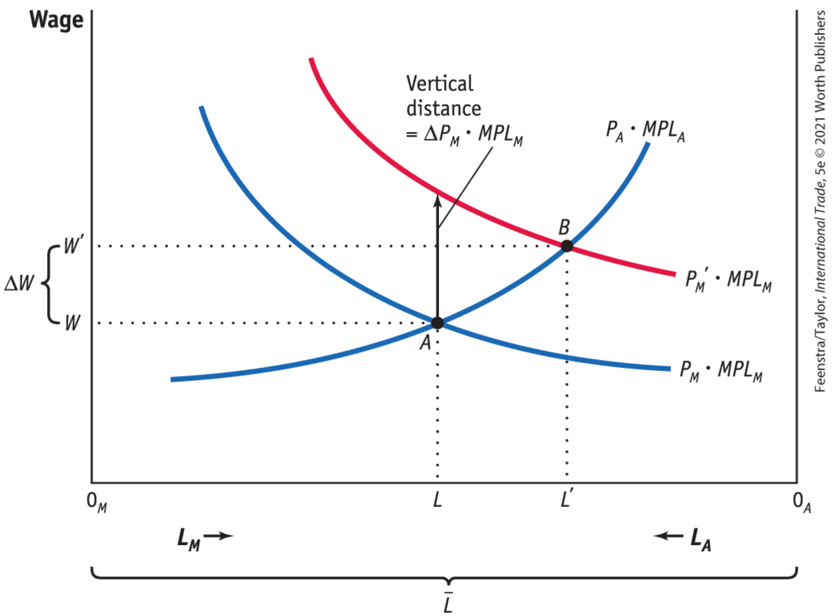
# 2) Earnings of Labor (part 5)

- Change in Relative Price of Manufactures
- Effect on Real Wages
  - As we can see from Figure 3-5, the increase in the wage from $W$ to $W'$ is *less than* the vertical increase $\Delta P_M \bullet MPL_M$
  - Since $\Delta W/W < \Delta P_M / P_M$, the percentage increase in the wage is *less than* the percentage increase in the price of the manufactured good.
  - This inequality means that the amount of the manufactured good that can be purchased with the wage has fallen.
  - Therefore, the *real wage in terms of the manufactured good* $W/P_M$ has decreased.

# 2) Earnings of Labor (part 6)

- Change in Relative Price of Manufactures
- Effect on Real Wages

FIGURE 3-5




Once again, since $\Delta W/W < \Delta P_M / P_M$, the percentage increase in the wage is *less than* the percentage increase in the price of the manufactured good.

The manufactured good that can be purchased with the wage has fallen.

Therefore, the *real wage in terms of the manufactured good W/PM* has decreased.

# 2) Earnings of Labor (part 7)

- Change in Relative Price of Manufactures
- Overall Impact on Labor
  - In the specific-factors model, the increase in the price of the manufactured good has an ambiguous effect on the real wage and therefore an ambiguous effect on the well-being of workers. Although ambiguous, this conclusion is important.
  - The result is different from what was found in the Ricardian model, where labor unambiguously earned a higher real wage.
  - This warns us that one cannot make unqualified statements about the effects of trade on workers.
  - The effect of trade on real wages can be complex.

# 2) Earnings of Labor (part 8)

- Change in Relative Price of Manufactures
- Unemployment in the Specific-Factors Model
  - It is hard to combine business cycle models with international trade models to isolate the effects of trade on workers.
  - Once we recognize that workers can find new jobs—possibly in export industries that are expanding—we still cannot conclude that trade is necessarily good or bad for workers.
  - Next we look at some evidence from the United States on the amount of time it takes to find new jobs and on the wages earned; we also look at attempts by governments to compensate workers who lose their jobs because of import competition. This type of compensation is called **Trade Adjustment Assistance (TAA)** in the United States.

# 3) Earnings of Capital and Land (part 1)

- Determining the Payments to Capital and Land
- If $Q_M$ is the output in manufacturing and $Q_A$ is the output in agriculture, the revenue earned in each industry is $P_M \bullet Q_M$ and $P_A \bullet Q_A$, and the payments to capital and to land are:

- $$\text{Payment to capital} = P_M \cdot Q_M - W \cdot L_M$$

$$\text{Payments to land} = P_A \cdot Q_A - W \cdot L_A$$

# 3) Earnings of Capital and Land (part 2)

- Determining the Payments to Capital and Land
- The earnings of one unit of capital (a machine, for instance), which we call $R_K$, and the earnings of an acre of land, which we call $R_T$, are calculated as:

$$R_K = \frac{\text{Payments to capital}}{K} = \frac{P_M \cdot Q_M - W \cdot L_M}{K}$$

$$R_T = \frac{\text{Payments to land}}{T} = \frac{P_A \cdot Q_A - W \cdot L_A}{T}$$

- Economists call $R_K$ the rental on capital and $R_T$ the rental on land.

# 3) Earnings of Capital and Land (part 3)

- Determining the Payments to Capital and Land
- Change in the Real Rental on Capital
  - As more labor is used in manufacturing, the marginal product of capital will rise because each machine has more labor to work it.
  - In addition, as labor leaves agriculture, the marginal product of land will fall because each acre of land has fewer laborers to work it.
  - The general conclusion is that *an increase in the quantity of labor used in an industry will raise the marginal product of the factor specific to that industry, and a decrease in labor will lower the marginal product of the specific factor*.

# 3) Earnings of Capital and Land (part 4)

- Determining the Payments to Capital and Land
- With labor leaving agriculture, the marginal product of each acre falls, so $R_T/P_A$ also falls.
- The fact that $R_T\ /P_A$ falls means that the real rental on land in terms of food has gone down, so landowners cannot afford to buy as much food.
- Thus, landowners are clearly worse off from the rise in the price of the manufactured good because they can afford to buy less of both goods.

# 3) Earnings of Capital and Land (part 5)

- Determining the Payments to Capital and Land
- Summary
  - An increase in the relative price of an industry's output will increase the real rental earned by the factor specific to that industry but will decrease the real rental of factors specific to other industries.
  - This conclusion means that:
    - The specific factors used in export industries will generally gain as trade is opened.
    - The relative price of exports rises.
    - The specific factors used in import industries will generally lose as trade is opened and the relative price of imports falls.

# 3) Earnings of Capital and Land (part 6)

$$Manufacturing\text{: Sales revenue} = P_M \cdot Q_M = \$100$$
$$\text{Payments to labor} = W \cdot L_M = \$60$$
$$\text{Payments to capital} = R_K \cdot K = \$40$$
$$Agriculture\text{: Sales revenue} = P_A \cdot Q_A = \$100$$
$$\text{Payments to labor} = W \cdot L_A = \$50$$
$$\text{Payments to land} = R_T \cdot T = \$50$$
$$Manufacturing\text{: Percentage increase in price} = \Delta P_M / P_M = 10\%$$
$$Agriculture\text{: Percentage increase in price} = \Delta P_A / P_A = 0\%$$
$$Both\ industries\text{: Percentage increase in the wage} = \Delta W / W = 5\%$$

Change in the Rental on Capital

$$\Delta R_K = \frac{\Delta P_M \cdot Q_M - \Delta W \cdot L_M}{K}$$

$$\frac{\Delta R_K}{R_K} = \frac{(\Delta P_M / P_M) \cdot P_M \cdot Q_M - (\Delta W / W) \cdot W \cdot L_M}{R_K \cdot K}$$

$$\frac{\Delta R_K}{R_K} = \frac{(10\% \cdot 100 - 5\% \cdot 60)}{40} = 17.5\%$$

Change in the Rental on Land

$$\Delta R_T = \frac{0 \cdot Q_A - \Delta W \cdot L_A}{T}$$

$$\frac{\Delta R_T}{R_T} = -\frac{\Delta W}{W}\left(\frac{W \cdot L_A}{R_T \cdot T}\right)$$

$$\frac{\Delta R_T}{R_T} = -5\%\left(\frac{50}{50}\right) = -5\%$$

# 3) Earnings of Capital and Land (part 7)

General Equation for the Change in Factor Prices

These equations summarize the response of all three factor prices in the short run, when capital and land are specific to each sector but labor is mobile.

$$\underbrace{\Delta R_T/R_T < 0}_{\substack{\textit{Real rental} \\ \textit{on land falls}}} \underbrace{< \Delta W/W < \Delta P_M/}_{\substack{\textit{Change in the real} \\ \textit{wage is ambiguous}}} \underbrace{P_M < \Delta R_K/R_K}_{\substack{\textit{Real rental} \\ \textit{on capital rises}}}, \text{ for an increase in } P_M$$

$$\underbrace{\Delta R_K/R_K < \Delta P_M/}_{\substack{\text{Real rental} \\ \text{on capital falls}}} \underbrace{P_M < \Delta W/W < 0}_{\substack{\text{Change in the real} \\ \text{wage is ambiguous}}} \underbrace{< \Delta R_T/R_T}_{\substack{\text{Real rental} \\ \text{on land rises}}}, \text{ for a decrease in } P_M$$

$$\underbrace{\Delta R_K/R_K < 0}_{\substack{\textit{Real rental} \\ \textit{on capital falls}}} \underbrace{< \Delta W/W < \Delta P_A/}_{\substack{\textit{Change in the real} \\ \textit{wage is ambiguous}}} \underbrace{P_A < \Delta R_T/R_T}_{\substack{\textit{Real rental} \\ \textit{on land rises}}}, \text{ for an increase in } P_A$$

# Conclusions (part 1)

- As long as the relative price with international trade differs from the no-trade relative price, a country will gain from international trade.
- The change in relative prices due to the opening of trade creates winners and losers. Some factors of production gain in real terms, and other factors of production lose.
- The factor that is specific to the importing-competing industry will lose. That industry suffers a drop in its relative price due to international trade, which leads to a fall in the real rental on the specific factor in that industry.
- The specific factor in the export industry, whose relative price rises with the opening of trade, enjoys an increase in real rental.

# Conclusions (part 2)

- Labor is mobile between the two industries, which allows it to avoid such extreme changes in wages. Real wages rise in terms of one good but fall in terms of the other good, so we cannot tell whether workers are better off or worse off after a country opens to trade.
- In theory, the gains of individuals as a result of the opening of trade exceed the losses. That means, in principle, that the government could tax the winners and compensate the losers in such a way that everyone would be better off. This has proven to be very challenging in practice.